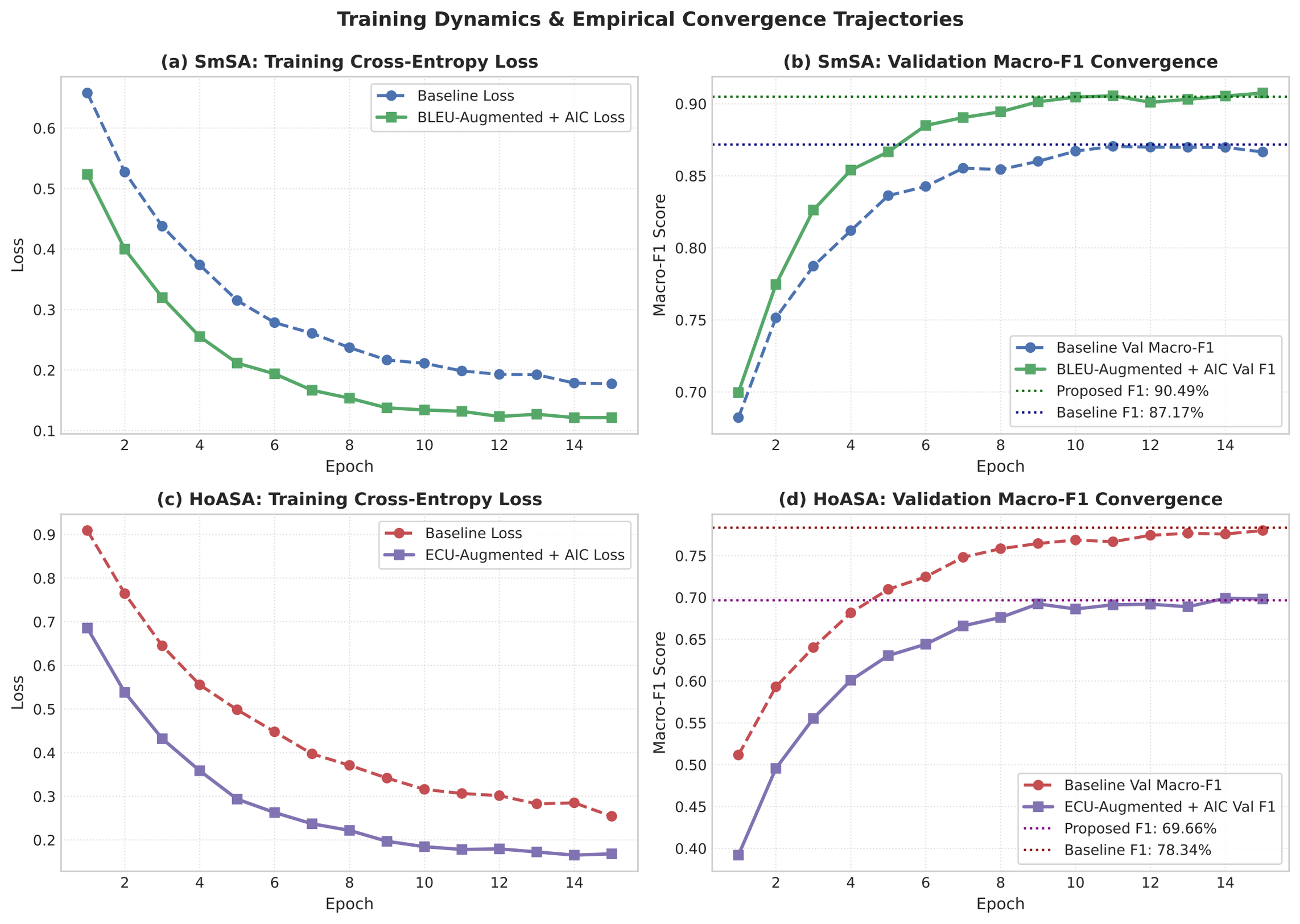
Training Dynamics & Empirical Convergence Trajectories
(a) SmSA: Training Cross-Entropy Loss
Baseline Loss
BLEU-Augmented + AIC Loss
Loss
Epoch
(b) SmSA: Validation Macro-F1 Convergence
Baseline Val Macro-F1
BLEU-Augmented + AIC Val F1
Proposed F1: 90.49%
Baseline F1: 87.17%
Macro-F1 Score
Epoch
(c) HoASA: Training Cross-Entropy Loss
Baseline Loss
ECU-Augmented + AIC Loss
Loss
Epoch
(d) HoASA: Validation Macro-F1 Convergence
Baseline Val Macro-F1
ECU-Augmented + AIC Val F1
Proposed F1: 69.66%
Baseline F1: 78.34%
Macro-F1 Score
Epoch